

Statistical Business Analysis Report

This project analyzes sales and customer churn datasets using statistical and business analytics techniques. The analysis includes descriptive statistics, data distribution analysis, correlation analysis, hypothesis testing, confidence intervals, and regression analysis to generate business insights.

1. Dataset Summary

Dataset	Rows	Purpose
Sales Data	100	Revenue and sales performance analysis
Customer Churn Data	500	Customer retention and churn analysis

2. Descriptive Statistics

The descriptive statistics were calculated on the Total Sales column to understand central tendency and variability.

Statistic	Value
Mean Sales	123,650.48
Median Sales	97,955.50
Mode	6,540
Standard Deviation	100,161.09
Minimum Sales	6,540
Maximum Sales	490,592

3. Data Distribution Analysis

The Shapiro-Wilk test was used to test the normality of the sales data.

Hypotheses:

- H1: Sales data is normally distributed
- H2: Sales data is not normally distributed

Result:

- p-value = 0.00000126

Conclusion:

Since the p-value is less than 0.05, the null hypothesis is rejected.

Therefore, the sales data is not normally distributed.

4. Correlation Analysis

Pearson correlation analysis was performed to identify relationships between variables.

Correlation between Product Price and Total Sales:

- Correlation Coefficient = 0.646

Interpretation:

The analysis shows a moderately strong positive relationship between product pricing and total sales revenue.

5. Hypothesis Testing

Test 1: One Sample T-Test

Business Question: Are average sales significantly different from the target sales value of 150,000?

Result: p-value = 0.00988

Conclusion: Significant difference found.

Test 2: Independent T-Test

Business Question: Do East and West regions have significantly different sales?

Result: p-value = 0.04548

Conclusion: Regional sales differences are statistically significant.

Test 3: ANOVA Test

Business Question: Do all regions have significantly different average sales?

Result: p-value = 0.0972

Conclusion: No significant difference found overall across all regions.

6. Confidence Interval Analysis

95% confidence intervals were calculated for average sales.

Confidence Interval:

- Lower Bound = 103,776.35
- Upper Bound = 143,524.61

Interpretation:

There is 95% confidence that the true average sales value lies within this range.

7. Regression Analysis

Linear regression analysis was performed to predict Total Sales using Product Price.

Results:

- R-squared = 0.417
- Regression Coefficient = 4.65

Interpretation:

Approximately 41.7% of the variation in sales can be explained by product price.

8. Business Insights

- Premium-priced products contribute significantly to overall revenue.
- Regional sales performance varies across locations.
- Customer churn appears higher among customers with larger monthly charges.
- Sales variability indicates possible seasonal or product-based fluctuations.

9. Recommendations

- Focus marketing efforts on high-performing products.
- Improve retention strategies for customers with higher monthly charges.
- Optimize underperforming regions using targeted campaigns.

10. Conclusion

This project successfully applied statistical methods to business datasets to derive meaningful insights. The findings demonstrate the importance of statistical analysis in business decision-making and customer retention strategies.